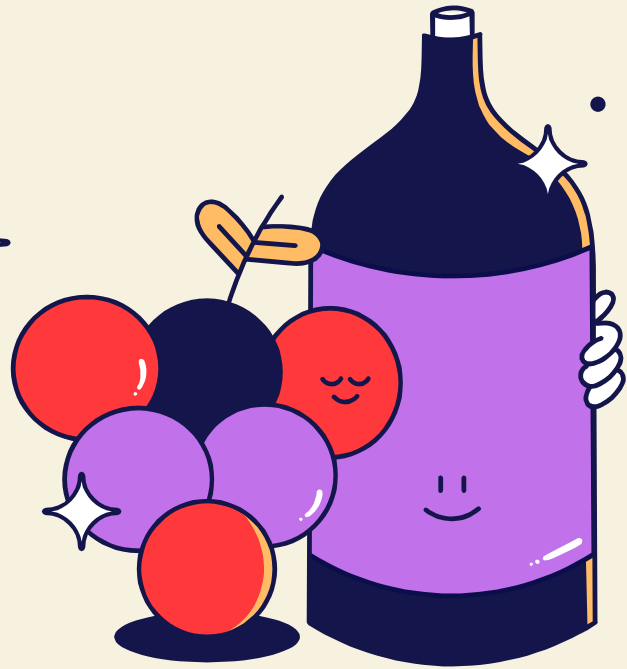


PREDICTING WINE HUE

Hannah Kim and Chris Tillotson

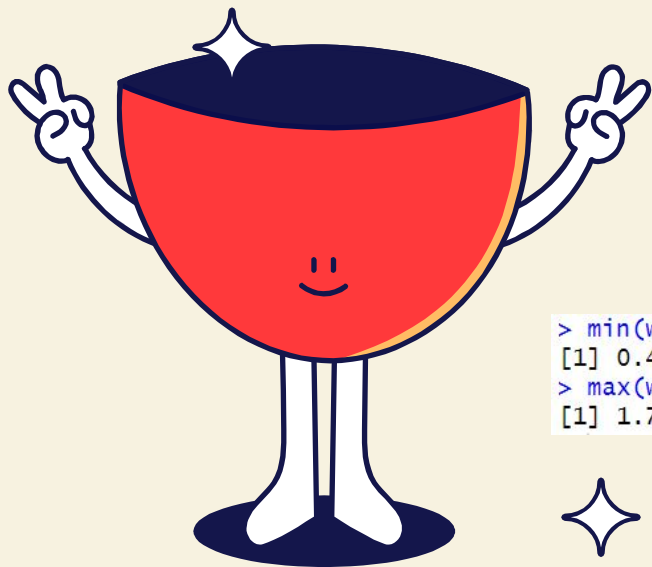


**The best
wine!**

OUR DATASET

- Set of 178 chemical samples of wine
- All samples from same region in Italy
- Derived from three different cultivars (Class)
 - 13 attributes measured
 - First 5 observations of dataset:

1	CLASS	ALCOHOL	MALIC_ACID	ASH	ALCALINITY_OF_ASH	MAGNESIUM	TOTAL_PHENOLS	FLAVANOIDS	NONFLAVANOID_PHENOLS	PROANTHOCYANINS	COLOR_INTENSITY	HUE	OD280/OD315_OF_DILUTED_WINES	PROLINE
2	CLASS 1	14.23	1.71	2.43	15.6	127	2.8	3.06	0.28	2.29	5.64	1.04	3.92	1065
3	CLASS 1	13.2	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.4	1050
4	CLASS 1	13.16	2.36	2.67	18.6	101	2.8	3.24	0.3	2.81	5.68	1.03	3.17	1185
5	CLASS 1	14.37	1.95	2.5	16.8	113	3.85	3.49	0.24	2.18	7.8	0.86	3.45	1480
6	CLASS 1	13.24	2.59	2.87	21	118	2.8	2.69	0.39	1.82	4.32	1.04	2.93	735



```
> min(wine$HUE)
[1] 0.48
> max(wine$HUE)
[1] 1.71
```

HUE

DEPENDENT VARIABLE

The tints found in the wine or the shade of red that the wine is

GOAL FOR ANALYSIS

Create a model to predict and estimate
the hue of wine based on a variety of
properties of the wine



FULL FIRST ORDER MODEL

HO: $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = \beta_8 = \beta_9 = \beta_{10} = \beta_{11} = \beta_{12} = \beta_{13} = \beta_{14} = 0$
 H1: at least one $\beta_i \neq 0$ for $i = 1-14$

$$\hat{Y} = 0.544 + 0.0705x_1 - 0.1215x_2 + 0.03088x_3 - 0.05321x_4 + 0.05176x_5 - 0.002829x_6 + 0.0004609x_7 + 0.0009344x_8 + 0.02428x_9 + 0.1784x_{10} - 0.006114x_{11} - 0.03503x_{12} - 0.00522x_{13} + 0.0001502x_{14}$$

```
> summary(model0)
```

Call:

```
lm(formula = HUE ~ CLASS + ALCOHOL + MALIC_ACID + ASH + ALCALINITY_OF_ASH +  
    MAGNESIUM + TOTAL_PHENOLS + FLAVANOIDS + NONFLAVANOID_PHENOLS +  
    PROANTHOCYANINS + COLOR_INTENSITY + OD280_OD315_OF_DILUTED_WINES +  
    PROLINE, data = wine)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.36212	-0.08762	-0.02171	0.08578	0.62793

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.440e-01	3.534e-01	1.539	0.125636
CLASSCLASS 2	7.050e-02	6.383e-02	1.105	0.270981
CLASSCLASS 3	-1.215e-01	8.926e-02	-1.361	0.175283
ALCOHOL	3.088e-02	2.307e-02	1.339	0.182514
MALIC_ACID	-5.321e-02	1.238e-02	-4.297	2.97e-05 ***
ASH	5.176e-02	6.329e-02	0.818	0.414647
ALCALINITY_OF_ASH	-2.829e-03	5.352e-03	-0.529	0.597786
MAGNESIUM	4.609e-04	9.112e-04	0.506	0.613633
TOTAL_PHENOLS	9.344e-04	3.694e-02	0.025	0.979851
FLAVANOIDS	2.428e-02	3.410e-02	0.712	0.477557
NONFLAVANOID_PHENOLS	1.784e-01	1.191e-01	1.499	0.135862
PROANTHOCYANINS	-6.114e-03	2.705e-02	-0.226	0.821487
COLOR_INTENSITY	-3.503e-02	8.931e-03	-3.922	0.000129 ***
OD280_OD315_OF_DILUTED_WINES	-5.220e-03	3.231e-02	-0.162	0.871834
PROLINE	1.502e-04	6.962e-05	2.158	0.032417 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1455 on 163 degrees of freedom
 Multiple R-squared: 0.627, Adjusted R-squared: 0.595
 F-statistic: 19.57 on 14 and 163 DF, p-value: < 2.2e-16

EXPLORATION

- Ran backwards stepwise regression
 - Eliminated 8 variables

```
> max.model <- lm(HUE ~ CLASS + ALCOHOL + MALIC_ACID + ASH + ALCALINITY_OF_ASH + MAGNESIUM + TOTAL_PHENOLS + FLAVANOIDS + NONFLAVANOID_PHENOLS + PROANTHOCYANINS + COLOR_INTENSITY + PROTEIN_CONCENTRATION + PROLINE)  
> step(max.model, direction = 'backward')
```

```
Start: AIC=-670.91  
HUE ~ CLASS + ALCOHOL + MALIC_ACID + ASH + ALCALINITY_OF_ASH +  
MAGNESIUM + TOTAL_PHENOLS + FLAVANOIDS + NONFLAVANOID_PHENOLS +  
PROANTHOCYANINS + COLOR_INTENSITY + PROTEIN_CONCENTRATION +  
PROLINE
```

	Df	Sum of Sq	RSS	AIC
- TOTAL_PHENOLS	1	0.00016	3.4699	-672.90
- PROANTHOCYANINS	1	0.00043	3.4702	-672.89
- ALCALINITY_OF_ASH	1	0.00162	3.4714	-672.83
- PROTEIN_CONCENTRATION	1	0.00187	3.4716	-672.82
- MAGNESIUM	1	0.00545	3.4752	-672.63
- FLAVANOIDS	1	0.00722	3.4770	-672.54
- ASH	1	0.00791	3.4777	-672.51
- ALCOHOL	1	0.02537	3.4951	-671.61
<none>			3.4698	-670.91
- NONFLAVANOID_PHENOLS	1	0.05008	3.5198	-670.36
- PROLINE	1	0.07272	3.5425	-669.22
- CLASS	2	0.15461	3.6244	-667.15
- COLOR_INTENSITY	1	0.32553	3.7953	-656.95
- MALIC_ACID	1	0.40758	3.8773	-653.14

```
Step: AIC=-683.67  
HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +  
PROLINE
```

	Df	Sum of Sq	RSS	AIC
<none>			3.5336	-683.67
- NONFLAVANOID_PHENOLS	1	0.05762	3.5912	-682.79
- PROLINE	1	0.10509	3.6387	-680.45
- COLOR_INTENSITY	1	0.31069	3.8443	-670.67
- MALIC_ACID	1	0.40172	3.9353	-666.50
- CLASS	2	0.49426	4.0278	-664.36



INDEPENDENT VARIABLES

01

Class

Qualitative; 3 different cultivars of wine

02

Malic Acid

Affects taste of wine; too much = sour, too little = flat

03

Nonflavanoid Phenols

Compounds localised mainly in pulp of grapes

04

Color Intensity

Concentration of color or opaqueness of the wine

05

Proline

Affects sweetness, bitterness, and intensity of grape flavor



PRELIMINARY FIRST ORDER MODEL

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0$$

$$H_1: \text{at least one } \beta_i \neq 0 \text{ for } i = 1-6$$

$$\hat{Y} = 1.123 - 0.01269x_1 - 0.2149x_2 - 0.05262x_3 + 0.1664x_4 - 0.02903x_5 + 0.0001436x_6$$

Call:

```
lm(formula = HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS +  
    COLOR_INTENSITY + PROLINE, data = wine)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.34073	-0.09032	-0.01759	0.09865	0.58947

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.123e+00	8.791e-02	12.773	< 2e-16 ***
CLASSCLASS 2	-1.269e-02	4.503e-02	-0.282	0.77836
CLASSCLASS 3	-2.149e-01	4.983e-02	-4.313	2.71e-05 ***
MALIC_ACID	-5.262e-02	1.193e-02	-4.409	1.83e-05 ***
NONFLAVANOID_PHENOLS	1.664e-01	9.967e-02	1.670	0.09677 .
COLOR_INTENSITY	-2.903e-02	7.486e-03	-3.878	0.00015 ***
PROLINE	1.436e-04	6.366e-05	2.255	0.02540 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1438 on 171 degrees of freedom
Multiple R-squared: 0.6179, Adjusted R-squared: 0.6045
F-statistic: 46.08 on 6 and 171 DF, p-value: < 2.2e-16

FIRST ORDER MODEL ANALYSIS



**Res. Std.
Error**

0.1438

Model will predict hue
within about $\pm 2s = 0.2876$
of its actual value

P-Value

$< 2.26 * 10^{-16}$

Less than $\alpha = 0.05$
Reject H_0
Model is useful for
prediction

Adj. R^2

0.6045

About 60% of sample
variation in hue is explained
by model



Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.012e-01	1.911e+00	0.315	0.754
CLASSCLASS 2	7.577e-01	1.993e+00	0.382	0.703
CLASSCLASS 3	-3.247e-01	2.202e+00	-0.147	0.883
MALIC_ACID	1.972e-01	6.098e-01	0.323	0.747
NONFLAVANOID_PHENOLS	2.374e+00	5.229e+00	0.454	0.650
COLOR_INTENSITY	-1.008e-01	2.430e-01	-0.415	0.679
PROLINE	3.524e-04	1.525e-03	0.231	0.818
I(MALIC_ACID^2)	6.681e-03	5.841e-02	0.114	0.909
I(NONFLAVANOID_PHENOLS^2)	8.058e-01	3.954e+00	0.204	0.839
I(COLOR_INTENSITY^2)	2.558e-03	1.597e-02	0.160	0.873
I(PROLINE^2)	-4.198e-08	6.221e-07	-0.067	0.946
CLASSCLASS 2:MALIC_ACID	-3.626e-01	6.304e-01	-0.575	0.566
CLASSCLASS 3:MALIC_ACID	-1.171e-01	6.344e-01	-0.185	0.854
CLASSCLASS 2:NONFLAVANOID_PHENOLS	-2.966e+00	5.386e+00	-0.551	0.583
CLASSCLASS 3:NONFLAVANOID_PHENOLS	-1.806e+00	5.428e+00	-0.333	0.740
CLASSCLASS 2:COLOR_INTENSITY	1.940e-01	2.739e-01	0.708	0.480
CLASSCLASS 3:COLOR_INTENSITY	-2.963e-02	2.703e-01	-0.110	0.913
CLASSCLASS 2:PROLINE	-7.179e-04	1.765e-03	-0.407	0.685
CLASSCLASS 3:PROLINE	1.857e-03	2.704e-03	0.687	0.493
CLASSCLASS 2:I(MALIC_ACID^2)	-6.854e-03	6.018e-02	-0.114	0.909
CLASSCLASS 3:I(MALIC_ACID^2)	-2.592e-02	6.060e-02	-0.428	0.670
CLASSCLASS 2:I(NONFLAVANOID_PHENOLS^2)	2.151e-01	4.089e+00	0.053	0.958
CLASSCLASS 3:I(NONFLAVANOID_PHENOLS^2)	1.086e-01	4.243e+00	0.026	0.980
CLASSCLASS 2:I(COLOR_INTENSITY^2)	3.832e-03	2.215e-02	0.173	0.863
CLASSCLASS 3:I(COLOR_INTENSITY^2)	5.067e-04	1.681e-02	0.030	0.976
CLASSCLASS 2:I(PROLINE^2)	1.275e-07	8.685e-07	0.147	0.884
CLASSCLASS 3:I(PROLINE^2)	-2.270e-06	1.688e-06	-1.345	0.181
MALIC_ACID:NONFLAVANOID_PHENOLS	-7.957e-01	8.134e-01	-0.978	0.330
MALIC_ACID:COLOR_INTENSITY	-1.795e-02	6.883e-02	-0.261	0.795
MALIC_ACID:PROLINE	1.085e-05	2.156e-04	0.050	0.960
NONFLAVANOID_PHENOLS:COLOR_INTENSITY	8.801e-02	4.472e-01	0.197	0.844
NONFLAVANOID_PHENOLS:PROLINE	-1.253e-03	2.325e-03	-0.539	0.591
COLOR_INTENSITY:PROLINE	4.910e-05	1.571e-04	0.313	0.755
CLASSCLASS 2:MALIC_ACID:NONFLAVANOID_PHENOLS	9.593e-01	8.331e-01	1.151	0.252
CLASSCLASS 3:MALIC_ACID:NONFLAVANOID_PHENOLS	8.767e-01	8.342e-01	1.051	0.295
CLASSCLASS 2:MALIC_ACID:COLOR_INTENSITY	1.637e-02	7.485e-02	0.219	0.827
CLASSCLASS 3:MALIC_ACID:COLOR_INTENSITY	1.706e-02	6.984e-02	0.244	0.807
CLASSCLASS 2:MALIC_ACID:PROLINE	6.037e-05	2.891e-04	0.209	0.835
CLASSCLASS 3:MALIC_ACID:PROLINE	3.130e-06	3.110e-04	0.010	0.992
CLASSCLASS 2:NONFLAVANOID_PHENOLS:COLOR_INTENSITY	-4.337e-01	4.944e-01	-0.877	0.382
CLASSCLASS 3:NONFLAVANOID_PHENOLS:COLOR_INTENSITY	-2.053e-01	4.568e-01	-0.449	0.654
CLASSCLASS 2:NONFLAVANOID_PHENOLS:PROLINE	2.506e-03	2.552e-03	0.982	0.328
CLASSCLASS 3:NONFLAVANOID_PHENOLS:PROLINE	1.846e-04	2.999e-03	0.062	0.951
CLASSCLASS 2:COLOR_INTENSITY:PROLINE	-1.292e-04	2.004e-04	-0.645	0.520
CLASSCLASS 3:COLOR_INTENSITY:PROLINE	1.265e-04	1.864e-04	0.679	0.498

FULL SECOND ORDER MODEL

$$H_0: \beta_1 = \dots = \beta_{44} = 0$$

$$H_1: \text{at least one } \beta_i \neq 0 \text{ for } i = 1-44$$

Residual standard error: 0.1466 on 133 degrees of freedom
 Multiple R-squared: 0.691, Adjusted R-squared: 0.5888
 F-statistic: 6.761 on 44 and 133 DF, p-value: < 2.2e-16

FULL SECOND ORDER MODEL ANALYSIS



**Res. Std.
Error**

0.1466

Model will predict hue
within about $\pm 2s = 0.2932$
of its actual value

P-Value

$< 2.26 * 10^{-16}$

Less than $\alpha = 0.05$
Reject H_0
Model is useful for
prediction

Adj. R^2

0.5888

About 59% of sample
variation in hue is explained
by model



NESTED F - TEST

```
Model 1: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +  
PROLINE + I(MALIC_ACID^2) + I(NONFLAVANOID_PHENOLS^2) + I(COLOR_INTENSITY^2) +  
I(PROLINE^2) + CLASS * MALIC_ACID + CLASS * NONFLAVANOID_PHENOLS +  
CLASS * COLOR_INTENSITY + CLASS * PROLINE + CLASS * I(MALIC_ACID^2) +  
CLASS * I(NONFLAVANOID_PHENOLS^2) + CLASS * I(COLOR_INTENSITY^2) +  
CLASS * I(PROLINE^2) + MALIC_ACID * NONFLAVANOID_PHENOLS +  
MALIC_ACID * COLOR_INTENSITY + MALIC_ACID * PROLINE + NONFLAVANOID_PHENOLS *  
COLOR_INTENSITY + NONFLAVANOID_PHENOLS * PROLINE + COLOR_INTENSITY *  
PROLINE + CLASS * MALIC_ACID * NONFLAVANOID_PHENOLS + CLASS *  
MALIC_ACID * COLOR_INTENSITY + CLASS * MALIC_ACID * PROLINE +  
CLASS * NONFLAVANOID_PHENOLS * COLOR_INTENSITY + CLASS *  
NONFLAVANOID_PHENOLS * PROLINE + CLASS * COLOR_INTENSITY *  
PROLINE  
Model 2: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +  
PROLINE  
Res.Df    RSS    Df Sum of Sq    F Pr(>F)  
1    133 2.8571  
2    171 3.5336 -38    -0.6765 0.8287 0.745
```

$$0.745 > 0.05$$

Stay with first order / more
parsimonious model - however, we
still want to check different reduced
versions of the second order model



Wine
party!

CHECK MORE MODELS

Model 4

HO: $\beta_1 = \dots = \beta_{32} = 0$
H1: at least one $\beta_i \neq 0$ for $i = 1-32$

Residual standard error: 0.1423 on 145 degrees of freedom
Multiple R-squared: 0.6826, Adjusted R-squared: 0.6125
F-statistic: 9.744 on 32 and 145 DF, p-value: $< 2.2e-16$

Remove Qualitative *
Quantitative terms

HO: $\beta_1 = \dots = \beta_{10} = 0$
H1: at least one $\beta_i \neq 0$ for $i = 1-10$

Residual standard error: 0.1443 on 167 degrees of freedom
Multiple R-squared: 0.6239, Adjusted R-squared: 0.6014
F-statistic: 27.71 on 10 and 167 DF, p-value: $< 2.2e-16$

Model 3

Remove Qualitative *
2nd order Quantitative
interaction terms

Residual standard error: 0.1429 on 161 degrees of freedom
Multiple R-squared: 0.6445, Adjusted R-squared: 0.6091
F-statistic: 18.24 on 16 and 161 DF, p-value: $< 2.2e-16$

HO: $\beta_1 = \dots = \beta_{16} = 0$
H1: at least one $\beta_i \neq 0$ for $i = 1-16$

Model 5

Remove all (Quantitative
and qualitative)
interaction terms

NESTED F - TESTS

MODEL 3

```
> anova(model1, model3)
Analysis of Variance Table
```

```
Model 1: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +
PROLINE
Model 2: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +
PROLINE + I(MALIC_ACID^2) + I(NONFLAVANOID_PHENOLS^2) + I(COLOR_INTENSITY^2) +
I(PROLINE^2) + CLASS * MALIC_ACID + CLASS * NONFLAVANOID_PHENOLS +
CLASS * COLOR_INTENSITY + CLASS * PROLINE + CLASS * I(MALIC_ACID^2) +
CLASS * I(NONFLAVANOID_PHENOLS^2) + CLASS * I(COLOR_INTENSITY^2) +
CLASS * I(PROLINE^2) + MALIC_ACID * NONFLAVANOID_PHENOLS +
MALIC_ACID * COLOR_INTENSITY + MALIC_ACID * PROLINE + NONFLAVANOID_PHENOLS *
COLOR_INTENSITY + NONFLAVANOID_PHENOLS * PROLINE + COLOR_INTENSITY *
PROLINE
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	171	3.5350				
2	145	2.9353	26	0.59969	1.1394	0.3057

$0.3057 > 0.05$
Stay with first order /
more parsimonious
model

MODEL 4

$0.2876 > 0.05$

Stay with first order /
more parsimonious
model

```
> anova(model1, model4)
Analysis of Variance Table
```

```
Model 1: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +
PROLINE
Model 2: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +
PROLINE + I(MALIC_ACID^2) + I(NONFLAVANOID_PHENOLS^2) + I(COLOR_INTENSITY^2) +
I(PROLINE^2) + +MALIC_ACID * NONFLAVANOID_PHENOLS + MALIC_ACID *
COLOR_INTENSITY + MALIC_ACID * PROLINE + NONFLAVANOID_PHENOLS *
COLOR_INTENSITY + NONFLAVANOID_PHENOLS * PROLINE + COLOR_INTENSITY *
PROLINE
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	171	3.5350				
2	161	3.2877	10	0.24727	1.2109	0.2876

MODEL 5

```
> anova(model1, model5)
Analysis of Variance Table
```

```
Model 1: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +
PROLINE
Model 2: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +
PROLINE + I(MALIC_ACID^2) + I(NONFLAVANOID_PHENOLS^2) + I(COLOR_INTENSITY^2) +
I(PROLINE^2)
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	171	3.5336				
2	167	3.4776	4	0.055936	0.6715	0.6126

$0.6126 > 0.05$
Stay with first order /
more parsimonious
model

MODEL 6

First order, remove Nonflavanoid Phenols

H0: $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$
H1: at least one $\beta_i \neq 0$ for $i = 1-5$

$$\hat{Y} = 1.139 + 0.01699x_1 - 0.1824x_2 - 0.04975x_3 - 0.02837x_4 + 0.0001608x_5$$

```
> summary(model7)
```

Call:

```
lm(formula = HUE ~ CLASS + MALIC_ACID + COLOR_INTENSITY + PROLINE,  
    data = wine)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.33516	-0.09261	-0.01540	0.08529	0.61712

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.139e+00	8.718e-02	13.070	< 2e-16 ***
CLASSCLASS 2	1.699e-02	4.731e-02	0.359	0.719878
CLASSCLASS 3	-1.824e-01	4.886e-02	-3.733	0.000257 ***
MALIC_ACID	-4.975e-02	1.195e-02	-4.163	4.95e-05 ***
COLOR_INTENSITY	-2.837e-02	7.554e-03	-3.755	0.000237 ***
PROLINE	1.608e-04	6.654e-05	2.417	0.016684 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1444 on 172 degrees of freedom

Multiple R-squared: 0.6119, Adjusted R-squared: 0.6007

F-statistic: 54.24 on 5 and 172 DF, p-value: < 2.2e-16

NESTED F - TEST

```
> anova(model6,model11)  
Analysis of Variance Table
```

```
Model 1: HUE ~ CLASS + MALIC_ACID + COLOR_INTENSITY + PROLINE
```

```
Model 2: HUE ~ CLASS + MALIC_ACID + NONFLAVANOID_PHENOLS + COLOR_INTENSITY +  
          PROLINE
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	172	3.5886				
2	171	3.5350	1	0.053639	2.5947	0.1091

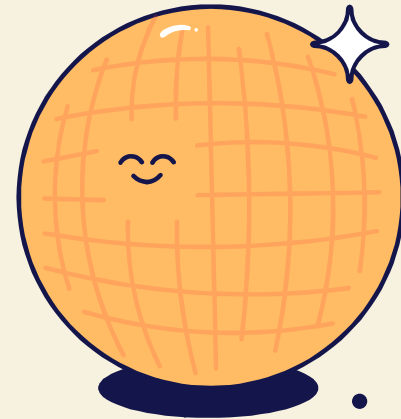
$$0.1091 > 0.05$$

Fail to reject H_0 , so we must go with
the more parsimonious model (model
6, without Nonflavanoid Phenols)



USE MODEL 6 AS OUR FINAL MODEL

Next Step: check that model meets
assumptions



CHECKING FOR MULTICOLLINEARITY

Variance Inflation Factor

```
> vif(model6)
```

	GVIF	Df	GVIF^(1/(2*Df))
CLASS	9.015644	2	1.732803
MALIC_ACID	1.510354	1	1.228965
COLOR_INTENSITY	2.579750	1	1.606160
PROLINE	3.437628	1	1.854084

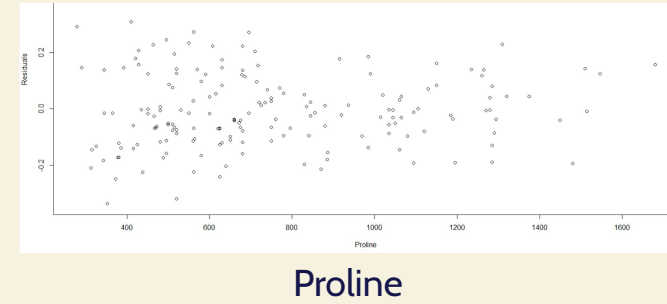
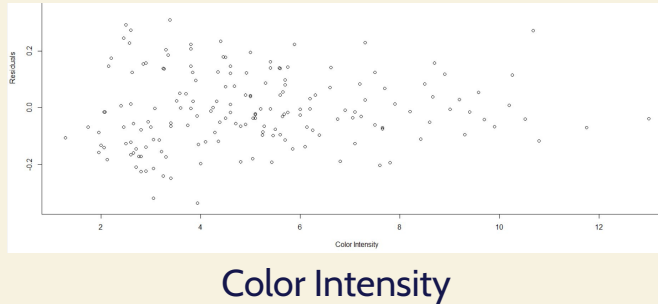
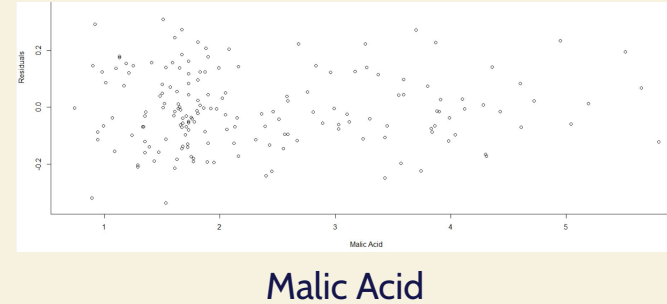
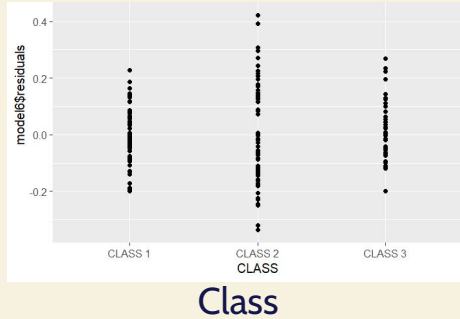
VIF < 10

Does not suggest
multicollinearity

Keep four predictor variables
in our model

RESIDUAL ANALYSIS

(Checking for homoscedasticity)



All predictor variables appear to have a homoscedastic variance, so transformations are unnecessary

F - Test for Heteroscedasticity



H0: True ratio of variances is equal to 1 (constant variance assumption is **not** violated)

H1: True ratio of variances is **not** equal to 1 (constant variance assumption is violated)



```
> var.test(model_first_half, model_second_half)
```

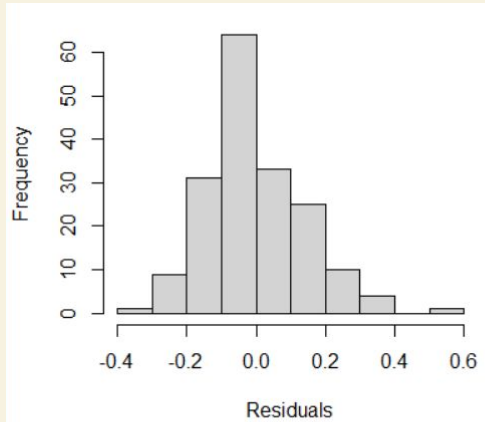
```
F test to compare two variances
```

```
data: model_first_half and model_second_half  
F = 0.76354, num df = 84, denom df = 84, p-value = 0.2184  
alternative hypothesis: true ratio of variances is not equal to 1  
95 percent confidence interval:  
 0.4963829 1.1744773  
sample estimates:  
ratio of variances  
 0.7635381
```

0.2184 > 0.05 , therefore we fail to reject H0 and conclude that the true ratio of variances is equal to 1 (constant variance assumption is **not** violated)

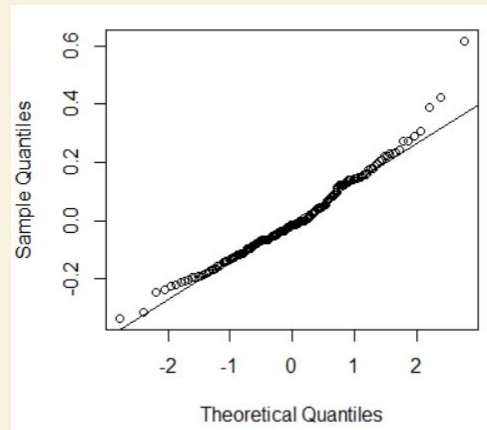


NORMALITY ASSUMPTION



Histogram of Residuals

Residuals are mound shaped



Quantile-Quantile Plot

Points mostly fall along the line

RESOURCES

Original Research Paper:

Stefan Aeberhard, Danny Coomans, Olivier de Vel, Comparative analysis of statistical pattern recognition methods in high dimensional settings, Pattern Recognition, Volume 27, Issue 8, 1994, Pages 1065-1077, ISSN 0031-3203,
[https://doi.org/10.1016/0031-3203\(94\)90145-7](https://doi.org/10.1016/0031-3203(94)90145-7).

UCI Machine Learning Repository Link:

<https://archive.ics.uci.edu/dataset/109/wine>

